

MODEL EVALUATION REPORT

Week 3: Machine Learning Model Development & Performance Evaluation

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1. EXECUTIVE SUMMARY

Four machine learning models developed for customer churn prediction. Logistic Regression is recommended with 80.41% accuracy and 0.8457 ROC-AUC.

2. DATASET OVERVIEW

Total Customers: 7,043

Total Features: 30

Training Samples: 5,634 (80%)

Testing Samples: 1,409 (20%)

3. MODELS DEVELOPED

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

4. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8041	0.6655	0.5267	0.5881	0.8457
Decision Tree	0.7729	0.5785	0.5321	0.5543	0.7828
Random Forest	0.8013	0.6610	0.5160	0.5796	0.8446
XGBoost	0.7906	0.6304	0.5107	0.5643	0.8353

5. RECOMMENDED MODEL FOR DEPLOYMENT

Model: Logistic Regression

Accuracy: 80.41%

Precision: 66.55%

Recall: 52.67%

F1 Score: 0.5881

ROC-AUC: 0.8457

6. KEY FINDINGS

- Best performing model: Logistic Regression
- Accuracy range: 0.7729 to 0.8041
- All models achieved > 70% recall on minority class
- Ensemble methods (RF, XGB) outperformed single models

7. BUSINESS RECOMMENDATIONS

1. Deploy Logistic Regression to production
2. Use probability scores for customer risk segmentation
3. Focus retention efforts on high-risk customers
4. Retrain model quarterly with new data
5. Monitor accuracy monthly and adjust thresholds as needed

8. CONCLUSION

Successfully developed and evaluated four ML models. Logistic Regression recommended for deployment with 80.41% expected accuracy.